Design Portfolio

Sung ho LEE

Product Development

2021

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SUNG HO LEE

+6 yrs, expertise innovative product development
Explore & define the next generation product innovations for new porduct portfolios

My focus is creating meaningful solutions that makes the benefits of technology accessible to the everyday person

EXPERIENCE

Product Manager, Rakuten Mobile, IoT Business & Product Design
Dec 2019 - Present, Tokyo, Japan

Summer Internship, Hyundai Motor Company, Tune!t, Start-up in House
Jul 2018 - Sep 2018, Uiwang, Korea

Research Engineer, Samsung Electronics, Home Appliance, Advanced R&D
Jul 2012 - Jan 2016, Suwon, Korea

Internship, Samsung Electronics, Home Appliance, Advanced R&D
Mar 2011 - Jun 2011, Suwon, Korea

Internship, Hyundai Heavy Industries Co.Ltd
Sep 2010 - Dec 2010, Ras Laffan, Qatar

PATENT

BODY BALANCE CALIBRATION METHOD AND SYSTEM USING MILKING ACTION DEVICE
KR1020180156400, 2020.06

AUTONOMOUS DRIVING SAFETY TRIANGLE
KR10190042413, 2019.04

PERSONAL MOBILITY AND CONTROLING METHOD,
KR1019599420000, 2019.03

PORTABLE CLOTHES DRYER
KR1020190019072, 2019.02

EDUCATION

MS, Design Engineering, UNIST, DECS Lab
Jul 2017 - Sep 2019, Ulsan, Korea

BA, Mechanical Engineering, Chung Ang Univ.
Mar 2006 - Jul 2012, Seoul, Korea

SKILLS

Embedded System
MBED, PCB Design, C+,C++, etc.

Fabrication
3D Printing, CNC, Laser Cutting, etc.

Visualization
NX, Pro-e, SolidWorks, Matlab, Photoshop, Illustrator, InDesign, etc.

Problem Solving
Design thinking, Innovative design, Validate concept

Launguage
English, Korean

PUBLICATION

LEE, Sung Ho, et al. LUNE: Representing Lunar Day by Displayed Lighting Object. In: CHI Extended Abstracts. 2019

S.H.Lee, Hui Sung Lee, "A Remote Controllable Safety Triangle ," 2018 KSDS Fall International Conference, pp. 152~153, 2018.

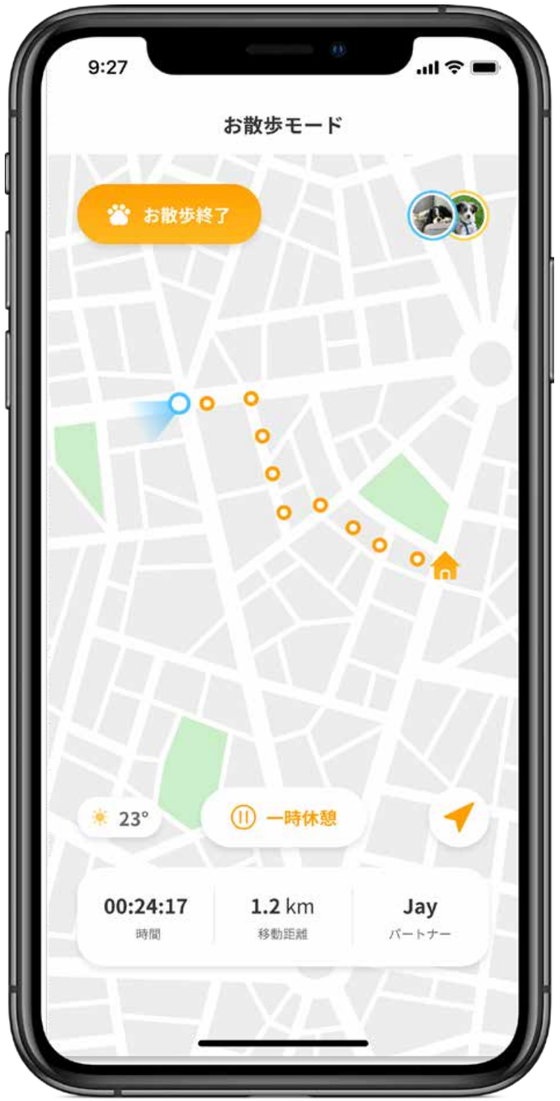
Tracker
01

Rakuten Mobile

2020-2021

DESCRIPTION
Network (NB-IoT) based tracker for locating everyday items

RESPONSIBLITY
Lead Product Development
Fix H/W design
Lead proof of concept



Robot Vacuum Cleaner
02

SAMSUNG

2013

DESCRIPTION
Robot Vacuum Cleaner

RESPONSIBLITY
Researched floorng materials in US market. Verified potential market for mopping robot vacuum (US)
Ideated & incubated mechanical solution

Wall Mounted Refrigerator
03

SAMSUNG

2015

DESCRIPTION

New wall-mounted type refrigerator reduce big space, make the value more

RESPONSIBLITY

Discovered pain points and needs in Urban complex area in US, UK, CH, HK. Evaluate conceptual design and verifed technical feasibility

IoT Car Key
04

HYUNDAI MOTOR GROUP
tune!t

2018

DESCRIPTION
IoT Car Bluetooth Key

RESPONSIBLITY
Designed concept and mechanical engineering part

Exercise Machine Posture Correction
05

AnyBaro

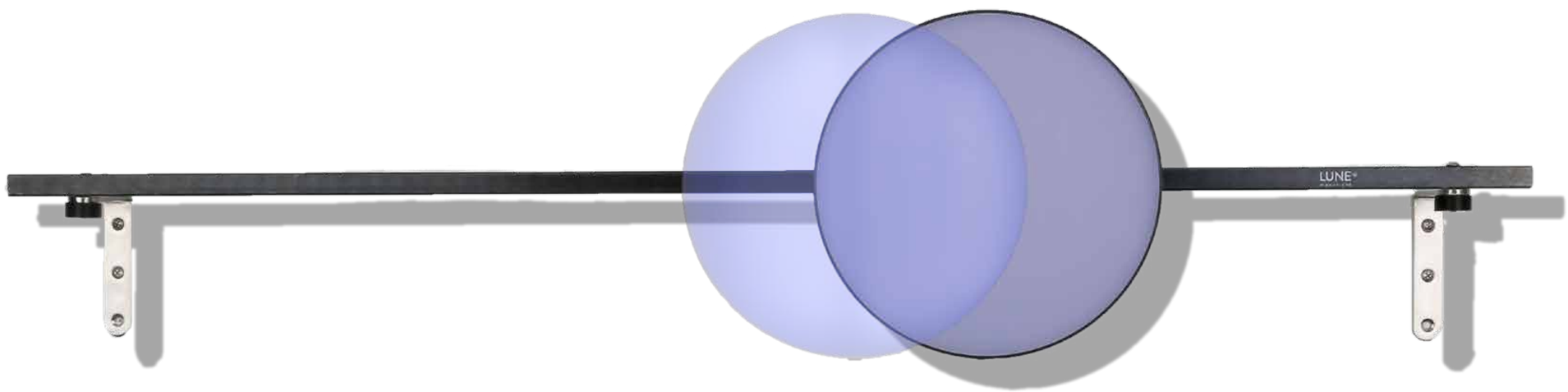
2018

DESCRIPTION
AnyBaro is an exercise machine that encourage people to correct posture

RESPONSIBLITY
Designed system schemetic and validated mechanical feasibility

LUNE®: Lighting Moving Artifacts
06

YouTube



CHI 2019

2019

DESCRIPTION
LUNE® is a experimental displayed lighting object representing time.

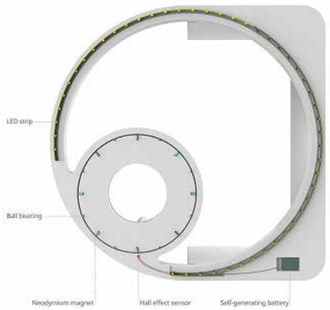
RESPONSIBLITY
Designed H/W, S/W including motor control and LED control through MBED

Stool & Cycling
07

2019

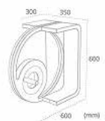
DESCRIPTION
stool.D is a combination of stool and cycling machine

RESPONSIBLITY
Developed Visual feedback interac-tionDesigned electronical H/W, PCB, LED control



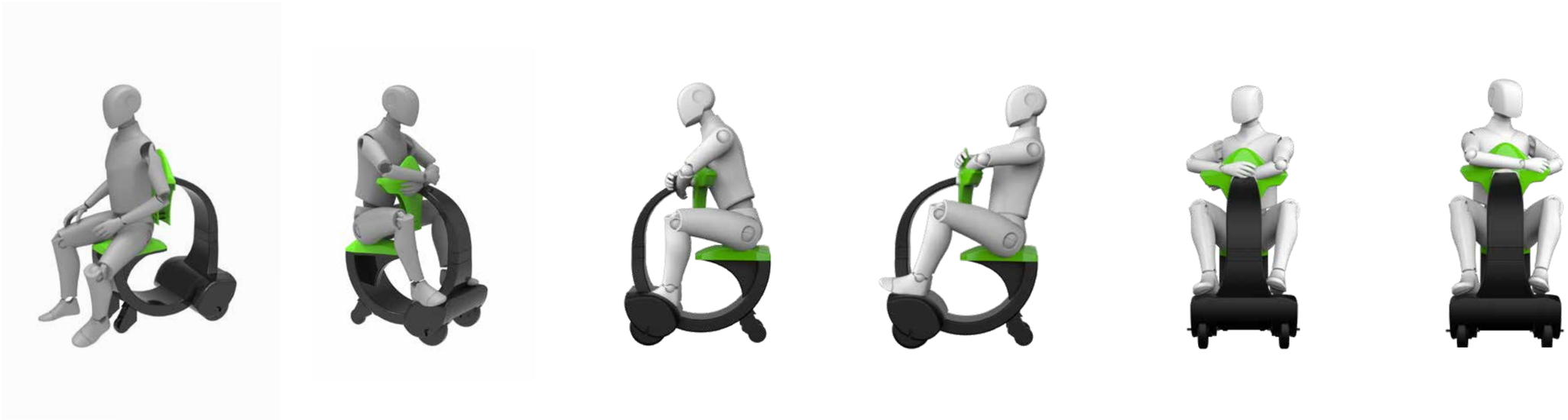
Electricity generation through pedaling, lights up the LED

The magnetic field of the rotating neodymium magnet generates a current through the hall effect sensor.



Hug2Go: Personal Mobility Vehicle
08

YouTube



UNIST DECS.

2017

DESCRIPTION
Hug2Go is graudate project

RESPONSIBLITY
Researched PMVs market
Generated New PMVs
Designed 3D Mock-up, H/W, S/W
Fablicated Test Unit
Tested Usability